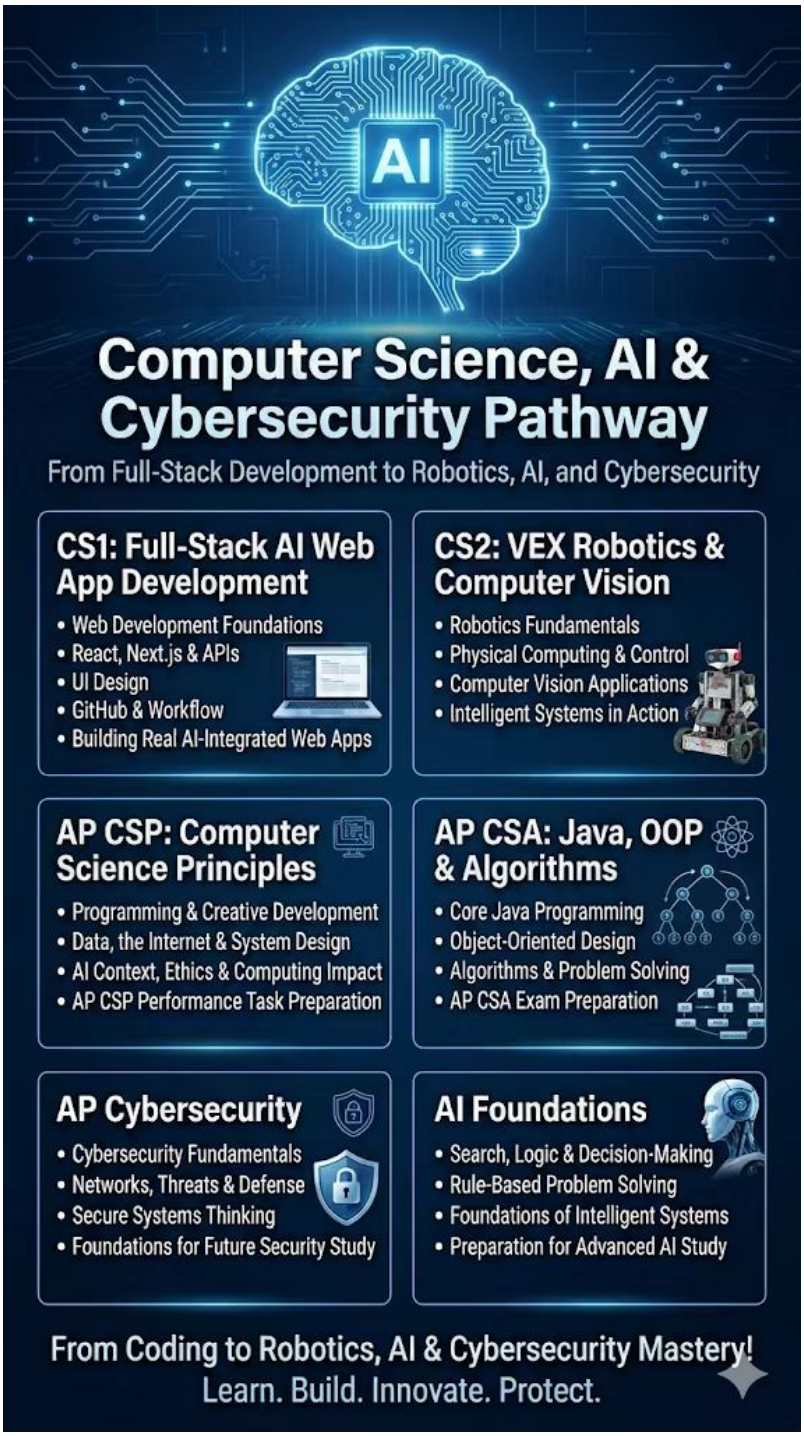
PATHWAY DESIGN & NEEDS ANALYSIS

# Computer Science, AI & Cybersecurity

From Full-Stack Development to Robotics, AI, and Cybersecurity

## Pathway Overview




## Design Rationale

### Software to Hardware Progression

This pathway is designed as a clear progression, not as a set of disconnected courses. First, students move from software to hardware. In CS1, they build strong foundations in programming, web development, APIs, and full-stack thinking. In CS2, they extend that knowledge into robotics, physical computing, and computer vision, so they begin to see how software connects to real-world systems.

### AI Embedded Throughout

AI is embedded throughout the pathway rather than being treated as a single isolated subject. AI-related ideas and applications are introduced across the first four courses, so students gradually develop familiarity with intelligent systems, computational thinking, and real-world AI contexts.

### AI Foundations as Capstone

AI Foundations is placed at the end as an advanced elective. By that stage, students have already built enough background in programming, systems, and problem solving. The final course then focuses more directly on common algorithms used across different areas of AI, helping students move beyond using AI tools toward understanding how AI works.

### AP-Aligned by Design

The pathway naturally covers AP CSP Big Ideas through hands-on practice before students formally take the AP course. For example, Big Idea 1 (Creative Development) maps directly to the complete software development cycle in CS1 — from UI design to full-stack deployment. Students enter AP CSP having already lived the concepts, not just studied them.

### In Short

**This pathway moves students from software, to hardware, to AI, with AI concepts integrated throughout the earlier courses, AP CSP Big Ideas covered through hands-on practice, and advanced AI algorithms taught in the final elective.**

## Course Prerequisites

| Course | Prerequisites |
|---|---|
| **Full-Stack Web & AI App Development** | Students should already be able to write and debug programs independently in at least one programming language including block coding. Not recommended for G6–G7 students; the course requires stronger math, English, and independent learning skills. Best for students who are ready to build real applications and learn through projects. |
| **Robotics & Intelligent System Development** | Students must already be able to build full-stack applications using Python and FastAPI. Includes robot status dashboard development and computer vision tasks. Not suitable for students without prior experience building applications with Python and FastAPI. Best for students interested in robotics, intelligent systems, and applied AI development. |
| **AP CSP** | Requires a MAP Reading RIT score of 220 or above. Requires an AP CSP mock exam score of 3 or above. Students should already be able to write and debug simple programs independently in any programming language. Best suited for students who can work independently and manage coursework on time. |
| **AP CSA** | Requires a MAP Reading RIT score of 220 or above. Requires an AP CSP exam score of 4 or above, or an AP CSA mock exam score of 3 or above. Students should already be able to write and debug programs independently in Java or another text-based programming language. Best suited for students who can work independently and manage coursework on time. |
| **AP Cybersecurity** | Requires a MAP Reading RIT score of 220 or above. Requires an AP CSP exam score of 4 or above. Students should be comfortable with computing concepts, network fundamentals, and logical reasoning. Best suited for students who can work independently and manage coursework on time. Completion of AP Computer Science A is strongly recommended. |
| **AI Foundations** | Students should demonstrate readiness in programming, code tracing, logical reasoning, and problem solving. Teacher recommendation is required. |

## $ Hardware Programming with VEX Robotics — Cost Breakdown

### First-Year Cost Estimate (10 Students)

- 5 VEX Competition Starter Kits needed (1 kit per 2 students)
- Cost per kit: **RMB 9,349**
- Total hardware cost: **RMB 46,745**
- Converted to USD: $46{,}745 \div 6.9056 \approx$ **USD 6,769.14**
- Teacher Cursor Pro+ plan: **USD 576 / year**
- 5 VEX V5 AI Vision Sensors: **USD 477.45**

**Total First-Year Cost: USD 7,822.59**